

9<sup>th</sup> Workshop on Algebraic Complexity Theory  
Debordering Closure Results in Determinantal and  
Pfaffian Ideals

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# What is Algebraic Complexity?

## Question

How hard is it to compute some explicit polynomial?

## Question

How hard it is to compute this polynomial *relative* to another polynomial?

# Algebraic Circuits with Oracles

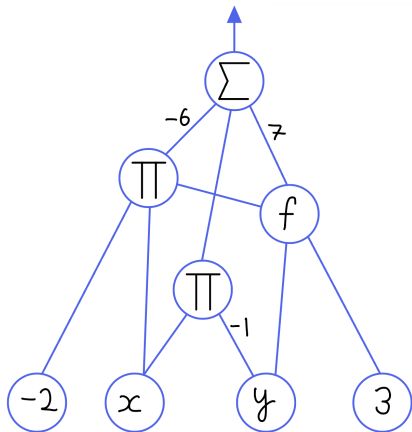


Figure:

$$-6(-2x \cdot f(y, 3)) - xy + 7f(y, 3)$$

## Question

How can we compare the *relative* complexity of two polynomials?

We have two measures of complexity:

- *Size*: The number of edges
- *Depth*: The length of the longest input  $\rightarrow$  output path

*Idea*: We are allowing a fixed  $f(\bar{x})$  to be computed with unit cost.

## Broadly Speaking

Theorem ([DG25])

Let  $X = (x_{i,j})$  be a matrix of indeterminates and let  $0 \neq f(X) \in \mathbb{F}[X]$  be a *“polynomial in the  $r \times r$  minors”* of  $X$ .

$$f(X) = \sum_{i \in I} h_i(X) g_i(X)$$

where each  $g_i(X)$  is an  $r \times r$  minor of  $X$  and  $h_i(X) \in \mathbb{F}[X]$ .

Then, we can use  $f(X)$  as an *oracle* to compute the  $t \times t$  determinant,  $t = \Theta(r^{1/3})$ , with a *polynomial size, constant depth* circuit.

## Circuit Complexity for Ideals

We want to characterize the circuit complexity of *families* of polynomials.

### Definition

Fix some polynomials  $g_1(\bar{x}), \dots, g_k(\bar{x}) \in \mathbb{F}[\bar{x}]$ .

The *ideal* generated by  $g_1(\bar{x}), \dots, g_k(\bar{x})$  is the set of combinations:

$$\langle g_1, \dots, g_k \rangle = \left\{ \sum_{i=1}^k h_i(\bar{x}) \cdot g_i(\bar{x}) \mid h_i(\bar{x}) \in \mathbb{F}[\bar{x}] \right\}.$$

### Question

Consider  $0 \neq f \in \langle g_1, \dots, g_k \rangle$ . How does the complexity of  $f$  compare to the complexity of the generators  $g_1, \dots, g_k$ ?

# Principal Ideals

## Example

The *principal ideals* are generated by a single polynomial  $g$ .

If  $f \in \langle g \rangle$ , then  $g$  is a *factor* of  $f$ .

## Question

Consider  $0 \neq f \in \langle g \rangle$ . Does  $g$  have a small  $f$ -oracle circuit?

## Principal Ideals

Conjecture ([Bür00, Conjecture 8.3])

If  $g$  is a factor of  $f \neq 0$ ,  $\text{size}(g) \leq \text{poly}(\text{size}(f), \deg(f))$ .

The conjecture as stated is still open.

However, it is known in a relaxed setting known as *border computation*.

## Border Complexity

Instead of exact computation, we can ask to *approximate* a polynomial.

Instead of working over a field  $\mathbb{F}$ , we take a parameter  $\varepsilon$ , allow our coefficients to be in  $\mathbb{F}(\varepsilon)$ , and let  $\varepsilon$  tend to 0.

### Definition

A circuit  $C$  *border computes*  $g(\bar{x})$  if it is defined over  $\mathbb{F}(\varepsilon)$  and computes

$$g(\bar{x}) + \varepsilon \tilde{g}(\bar{x}, \varepsilon), \quad \tilde{g}(\bar{x}, \varepsilon) \in \mathbb{F}[[\varepsilon]][\bar{x}].$$

We *cannot* set  $\varepsilon = 0$  since we may divide by  $\varepsilon$  in  $C$ .

# What does Border Computation Get Us?

## Question

What is the power of border computation?

Is it strictly necessary? Can we prove something in the border setting and *deborder* the result to get an exact computation?

## Principal Ideals

Conjecture ([Bür00, Conjecture 8.3])

If  $g$  is a factor of  $f$ ,  $\text{size}(g) \leq \text{poly}(\text{size}(f), \deg(f))$ .

Theorem ([Bür04, Theorem 1.3])

Over fields of characteristic 0,  $g$  can be *border computed* by a circuit of size  $\text{poly}(\text{size}(f), \deg(f))$ .

Question

Can we *deborder* this result, that is can we remove this  $\varepsilon$  approximation?

## Debordering in Other Settings

For a class  $\mathcal{C}$ ,  $\overline{\mathcal{C}}$  is the set of polynomials border-computed by  $\mathcal{C}$ .

- $\overline{\Sigma\Pi} = \Sigma\Pi$  and  $\overline{\Pi\Sigma} = \Pi\Sigma$  [DL25].
- Debordering results for depth-3 and depth-4 already would have strong implications [AV08, GKKS16].
- Debordering *Waring Rank* dates back to early work in classical algebraic geometry [Pal06, Ter11].

$$\text{Waring}(f) = \min_r \left\{ f(\overline{x}) = \sum_{i=1}^r \ell_i(\overline{x})^r \mid \deg(\ell_i) = 1 \right\}$$

- Forbes conjectured that if  $f \in \overline{\Sigma^s \wedge^d \Sigma}$  then  $f \in \Sigma^{s'} \wedge^d \Sigma$  for  $s' \in \text{poly}(s, n, d)$  [Shp16].

## What about Two Generators?

Consider an ideal with two generators  $\langle g_1(\bar{x}), g_2(\bar{x}) \rangle$ .

Say  $0 \neq f(\bar{x}) \in \langle g_1(\bar{x}), g_2(\bar{x}) \rangle$ :

$$f(\bar{x}) = h_1(\bar{x})g_1(\bar{x}) + h_2(\bar{x})g_2(\bar{x}).$$

### Question

Do  $g_1(\bar{x})$  or  $g_2(\bar{x})$  have a small  $f$ -oracle circuit?

This is open for *general* polynomials  $g_1(\bar{x}), g_2(\bar{x})$  [Gro20].

## Determinantal Ideals

Questions about oracle circuit complexity are open for general ideals with more than one generator. We can instead ask about ideals whose generators have *additional structure*.

### Example

Consider an  $n \times m$  matrix  $X = (x_{i,j})$  of variables. Let  $I_{n,m,r}^{\det}$  be the *determinantal ideal* generated by the  $r \times r$  minors of  $X$ .

## A Conjecture on Determinantal Ideals

### Example

Consider an  $n \times m$  matrix  $X = (x_{i,j})$  of variables. Let  $I_{n,m,r}^{\det}$  be the *determinantal ideal* generated by the  $r \times r$  minors of  $X$ .

### Conjecture ([Gro20, Conjecture 6.3])

Let  $f \in I_{n,n,n/2}^{\det}$  be a nonzero polynomial.

There is a constant depth  $f$ -oracle circuit of size  $\text{poly}(n)$  that computes the  $t \times t$  determinant,  $t = n^{\Theta(1)}$ .

*Upshot:* This means that the complexity of the determinant is a *lower bound* on the complexity of  $I_{n,m,r}^{\det}$ .

## What does the Oracle Give Us?

Conjecture ([Gro20, Conjecture 6.3])

Let  $f \in I_{n,n,n/2}^{\det}$  be a nonzero polynomial.

There is a constant depth  $f$ -oracle circuit of size  $\text{poly}(n)$  that computes the  $t \times t$  determinant,  $t = n^{\Theta(1)}$ .

- There exist polynomial size circuits for computing  $\det(X)$ .
- However, these circuits are *not* constant-depth.
- The oracle allows *constant depth* computation of the determinant.

## Prior Closure Results in Determinantal Ideals

Theorem ([AF22, Theorem 1.1])

Let  $f \in I_{n,m,r}^{\det}$  be a nonzero polynomial.

Then, there is a depth-three  $f$ -oracle circuit of size  $O(n^2 m^2)$  that *border computes* the  $t \times t$  determinant for some  $t = \Theta(r^{1/3})$ .

Question

Can we *deborder* this result, that is can we remove this  $\varepsilon$  approximation?

# Closure Results in Determinantal Ideals

Theorem ([DG25, Theorem 1.5])

Let  $f \in I_{n,m,r}^{\det}$  be a nonzero polynomial.

Then, there is a depth-three  $f$ -oracle circuit of size  $\text{poly}(n, m, \deg(f))$  that *exactly computes* the  $t \times t$  determinant for some  $t = \Theta(r^{1/3})$ .

Main Tools:

- Both [AF22, DG25] use *Straightening Laws* from Invariant Theory to express  $f(X)$  in a standard basis indexed by combinatorial objects.
- To get a circuit for a specific basis term, we use *Homogenization* as well as a new application of the *Isolation Lemma*.

## Proof Outline

1. Express  $f(X) \in I_{n,m,r}^{\det}$  in a combinatorial basis  $\mathcal{B}$ .
2. Give a linear change of variables  $x_{i,j} \mapsto \ell_{i,j}(X)$  so that

$$f(X) \mapsto \sum_{i \in I} K_i(X) + \text{lower degree "junk" terms,}$$

where  $K_i(X) \in \mathcal{B} \cap I_{n,m,r}^{\det}$ .

3. Use *homogenization* and *isolation* to get a circuit for a single  $K_i(X)$ .
4. Apply a projection  $K_i(X) \mapsto \det_t(Y)$ .

## Why Isolation and Homogenization?

1. Express  $f(X) \in I_{n,m,r}^{\det}$  in a combinatorial basis  $\mathcal{B}$ .
2. Give a linear change of variables  $x_{i,j} \mapsto \ell_{i,j}(X)$  so that

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3. Use *homogenization* and *isolation* to get a circuit for a single  $K_i(X)$ .
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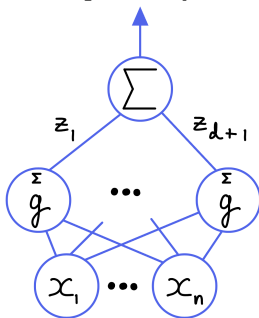
# Homogenization

## Definition

Consider a degree  $d$  polynomial  $g(\bar{x}, t) = \sum_{i=0}^d \text{coeff}_{t^i}(g)t^i$ .

## Lemma (Folklore)

Say  $g$  is computed by a size  $s$ , depth  $\Delta$   $f$ -oracle circuit with top  $\Sigma$ -gate.



Then, we can compute  $\text{coeff}_{t^i}(g)$  by a size  $O(\deg(g) \cdot s)$ , depth  $\Delta$   $f$ -oracle circuit.

## Issues with Homogenization

If a circuit border computes  $g(\overline{x})$ , then it computes

$$g(\overline{x}) + \varepsilon \tilde{g}(\overline{x}, \varepsilon), \quad \tilde{g}(\overline{x}, \varepsilon) \in \mathbb{F}[[\varepsilon]][\overline{x}].$$

*Idea:* Homogenize with respect to  $\varepsilon$ .

*Problem:*  $\deg_{\varepsilon}(\tilde{g})$  can be *arbitrarily large*

$\implies$  Homogenization gives *large circuit*.

## Idea: Kronecker Substitution

Say there is a specific coefficient  $c_{\bar{e}}$  in  $g(\bar{x}) = \sum_{\bar{e}} c_{\bar{e}} \bar{x}^{\bar{e}}$  we want to *isolate*.

### Lemma

Suppose that  $\deg(g) \leq d$ , then the *Kronecker substitution*

$$x_i \mapsto w^{d^i}$$

maps distinct monomials to distinct monomials.

*Problem:* The resulting polynomial has *large degree*.

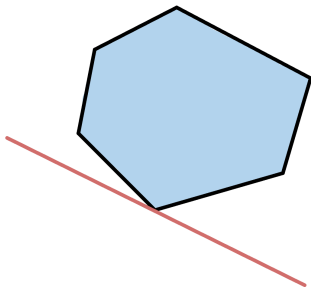
$\implies$  Homogenization gives *large circuit*.

## Isolation Lemma

Say there is a specific coefficient  $c_{\bar{e}}$  in  $g(\bar{x}) = \sum_{\bar{e}} c_{\bar{e}} \bar{x}^{\bar{e}}$  we want to *isolate*.

Lemma ([KS01, Lemma 4], [MVV87])

Linear programs with *random* cost functions will have a unique minimum.



Moreover, if the linear equations have bounded integer coefficients, then evaluation at *small, random* values has a unique minimum.

## Isolation Lemma

Lemma ([KS01, Lemma 4])

Let  $\mathcal{L}$  be any collection of distinct linear forms in variables  $z_1, \dots, z_\ell$  with coefficients in the range  $\{0, \dots, K\}$  for some integer  $K \in \mathbb{Z}_{\geq 0}$ . Let  $\varepsilon > 0$ .

Let  $z_1, \dots, z_\ell$  be independently and uniformly chosen from  $\{0, \dots, M\}$  at random, where  $M \geq K\ell/\varepsilon$ .

Then, with probability at least  $1 - \varepsilon$ , there is a unique form in  $\mathcal{L}$  of minimum value at  $z_1, \dots, z_\ell$ .

## Isolating Monomials

Lemma ([DG25, Corollary 2.27])

Consider a polynomial  $g(x_1, \dots, x_\ell)$  such that the individual degree of each  $x_i$  in  $g$  is at most  $K$ :

$$g(x_1, \dots, x_\ell) = \sum_{\bar{e} \in \mathcal{L}} c_{\bar{e}} x_1^{e_1} \cdots x_\ell^{e_\ell}.$$

Randomly choose  $z_1, \dots, z_\ell$  and define a morphism

$$x_i \mapsto w^{z_i}, \quad g \mapsto \sum_{\bar{e} \in \mathcal{L}} c_{\bar{e}} \cdot w^{\sum_{i=1}^{\ell} e_i z_i}.$$

The Isolation Lemma shows that the  $z_1, \dots, z_\ell$  can be choose to be small  
 $\implies$  unique lowest  $\deg_w$ -term and homogenization w.r.t  $w$  is small.

# Pfaffian Analogues

## Definition

If  $X$  is a  $2n \times 2n$  skew-symmetric matrix, then  $\det(X)$  is the perfect square of another polynomial called the *Pfaffian* of  $X$ .

## Theorem ([DG25, Theorem 1.6])

Let  $f \in I_{2n, 2r}^{\text{pfaff}}$  be a nonzero polynomial. Then there is a depth-three  $f$ -oracle circuit of size  $\text{poly}(n, \deg(f))$  that *exactly computes* the  $t \times t$  pfaffian for some  $t = \Theta(r^{1/3})$ .

## Symmetric Analogue?

The focus on determinantal and pfaffian ideals stem from *invariant theory*.  
Are there other natural settings to study from there?

### Question

Is there an analogue to our results for the ideal of determinants of a symmetric matrix?

## Roadblocks for the Permanent

The other big star in algebraic complexity is the *permanent* of a matrix.

### Question

Is there an analogue to our results for the ideal of permanents of a matrix?

The main roadblock is that while the determinant and Pfaffian are alternating in the rows, but the permanent is not.

# Thank You!

*Like the Arabian phoenix rising out of its ashes, the theory of invariants, pronounced dead at the turn of the century, is once again at the forefront of mathematics.*

— Joseph P.S. Kung, Gian-Carlo Rota [[KR84](#)]

Slides can be found on my site [anakin.phd](#)

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